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# ENGINEERING PORTFOLIO

AMULYA PATHANIA

BASc. IN MANUFACTURING ENGINEERING

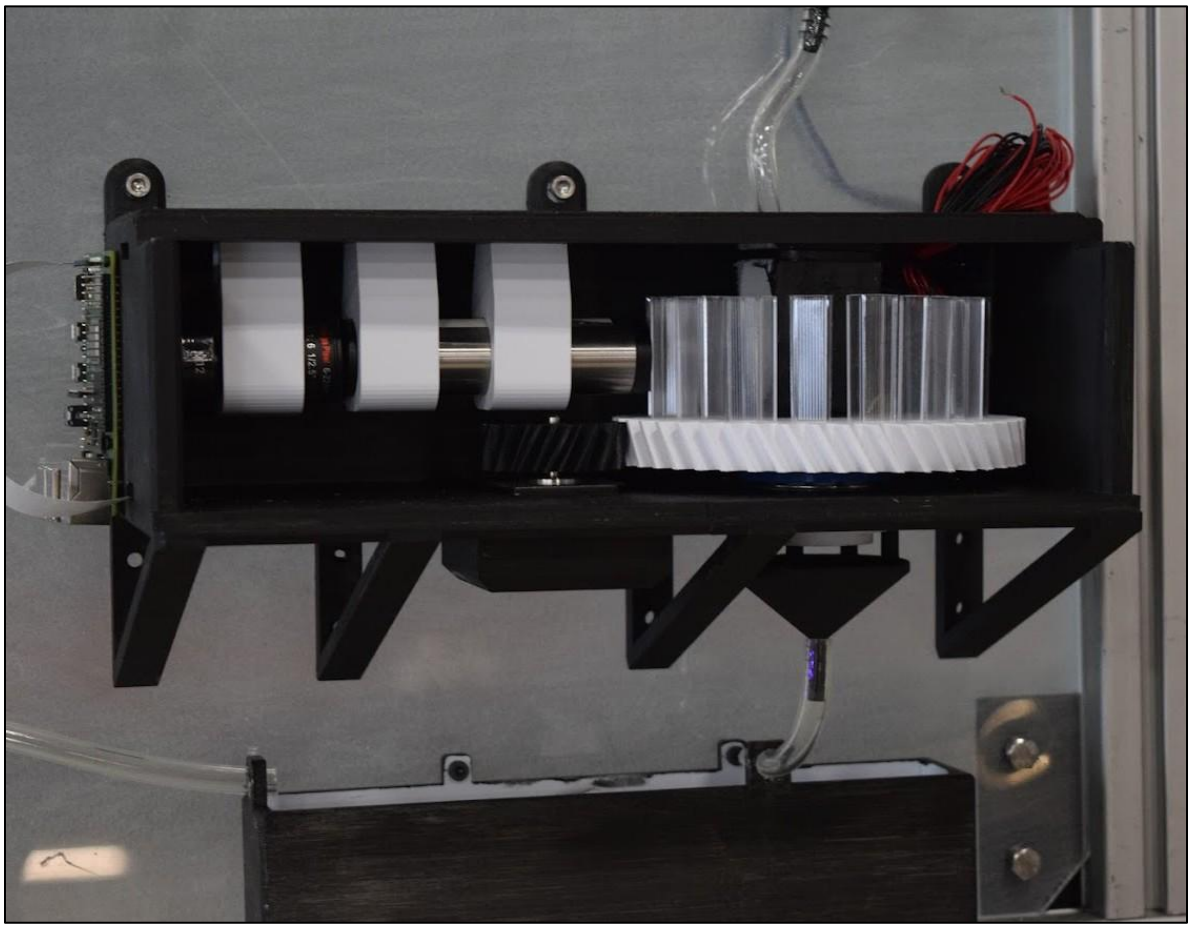


*Figure 1: Final 3D-Printed and Fully Assembled Carousel*

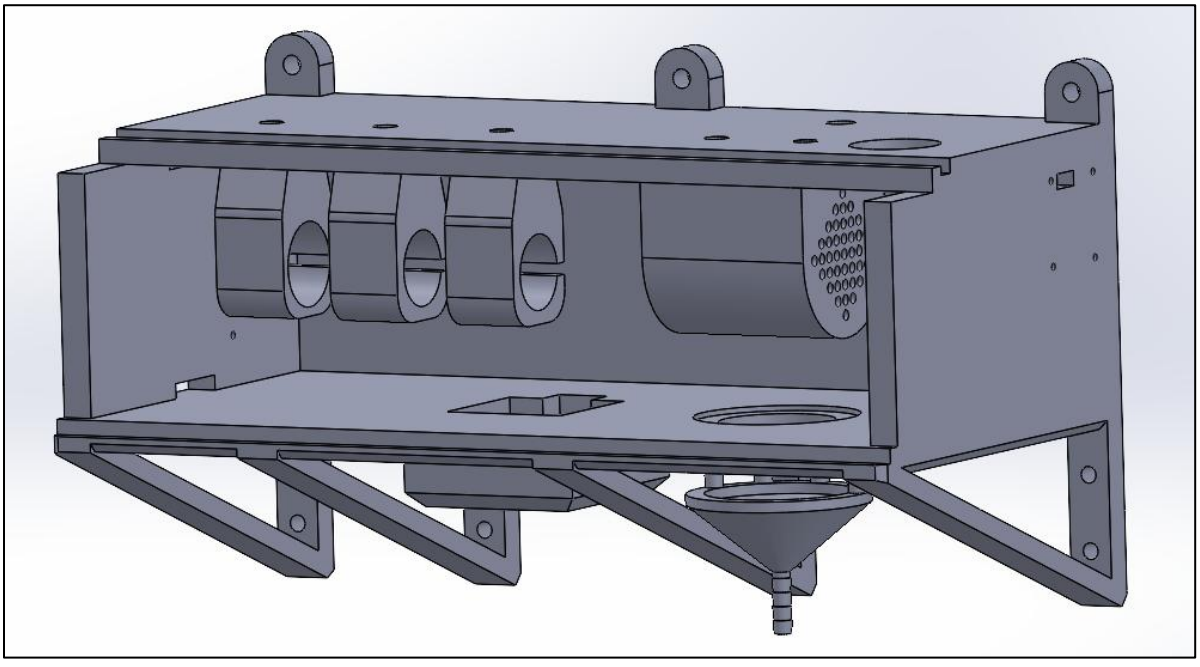


*Figure 2: SolidWorks Model Without Additional Components*

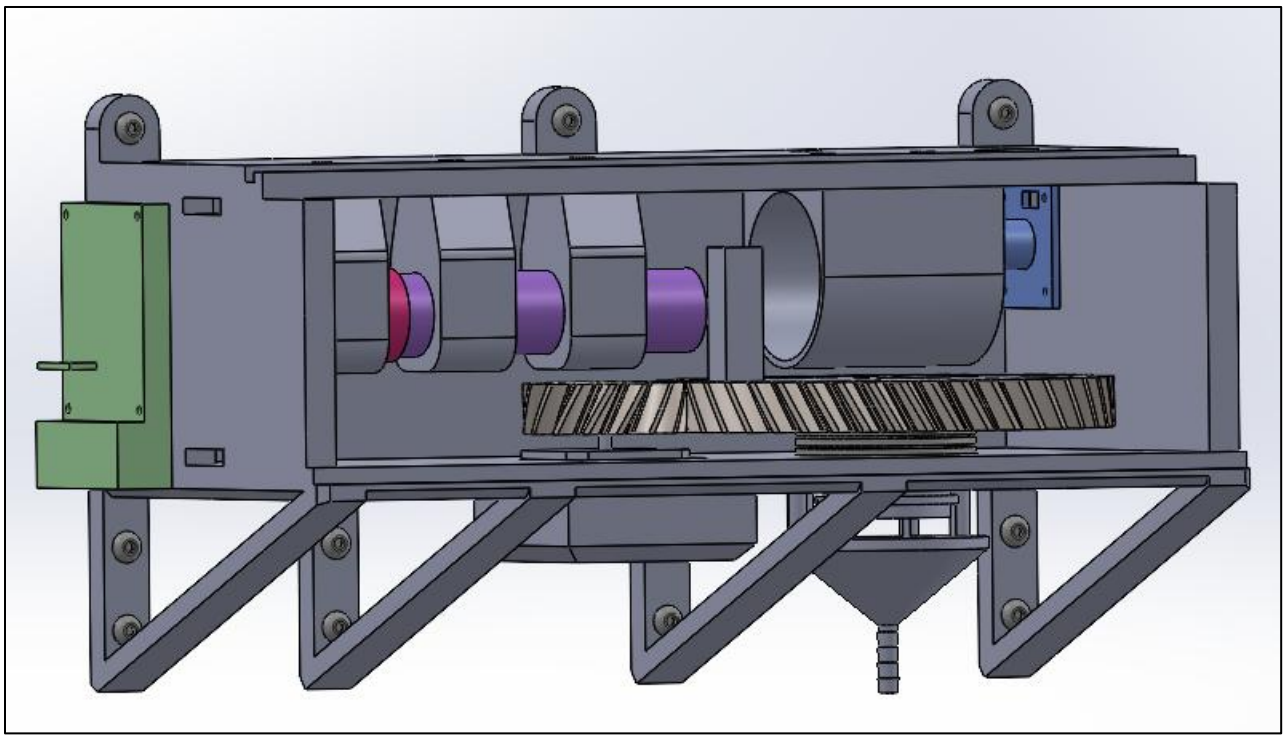


*Figure 3: Left Perspective View of the SolidWorks Model with Integrated Components*

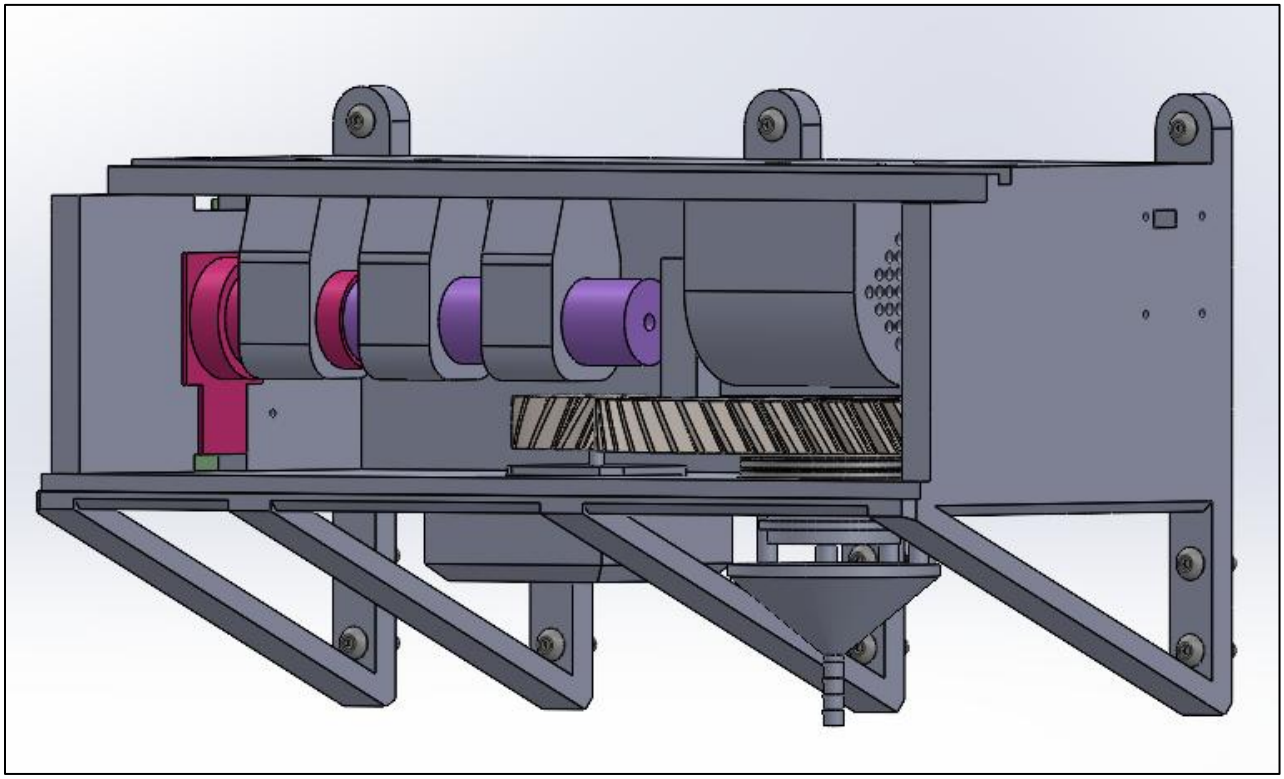


*Figure 4: Right Perspective View of the SolidWorks Model with Integrated Components*

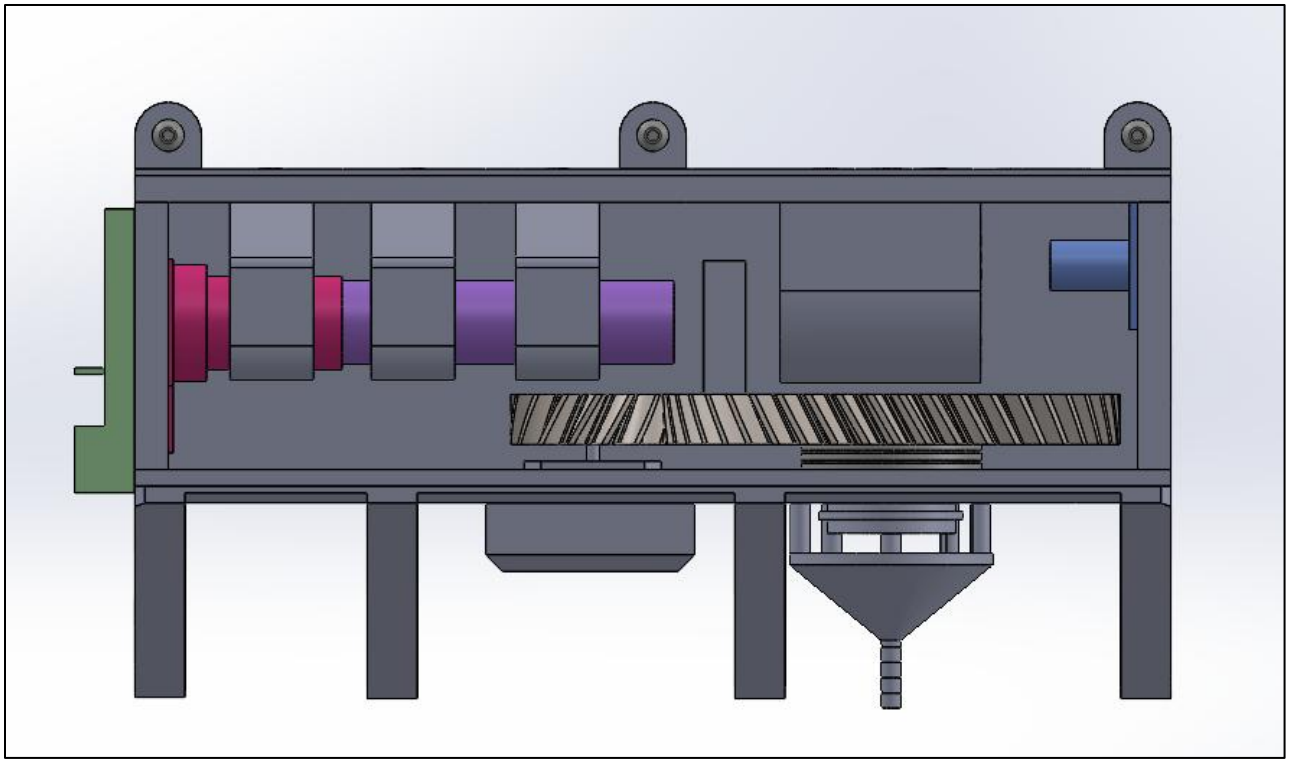


*Figure 5: Front View of the SolidWorks Model with Integrated Components*

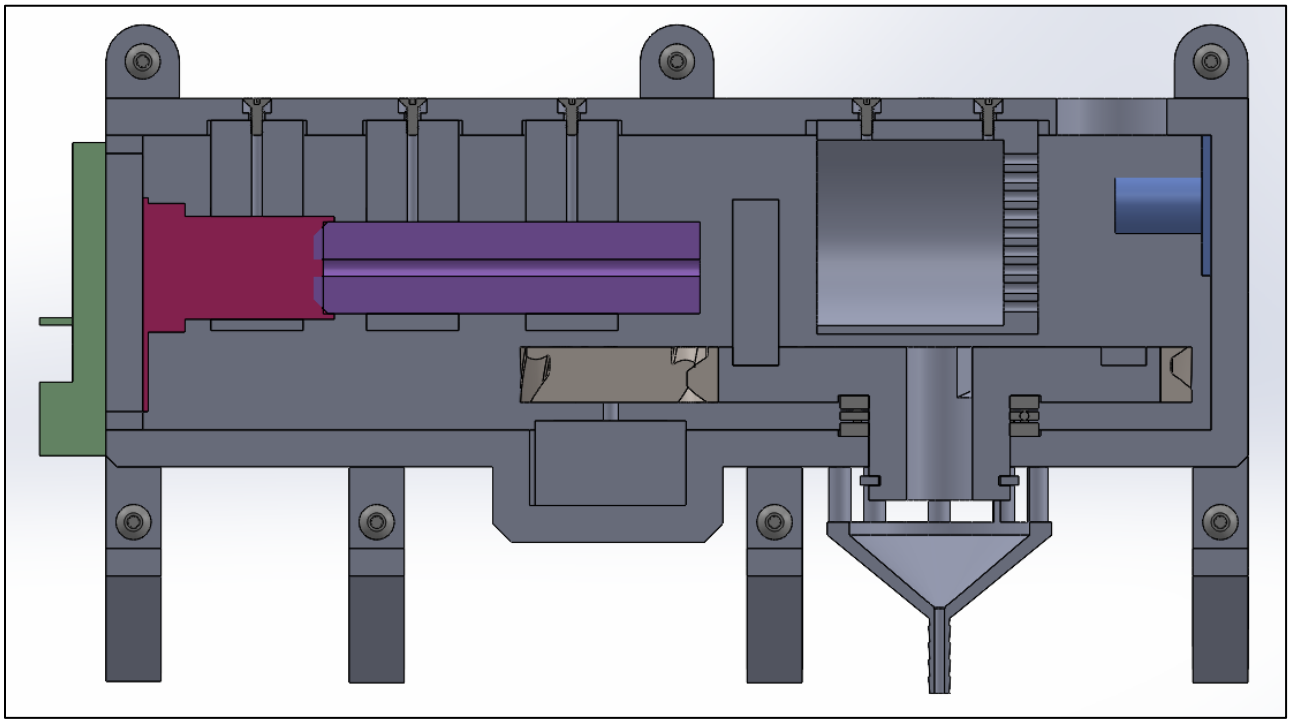


*Figure 6: Section View of the SolidWorks Model with Integrated Components*

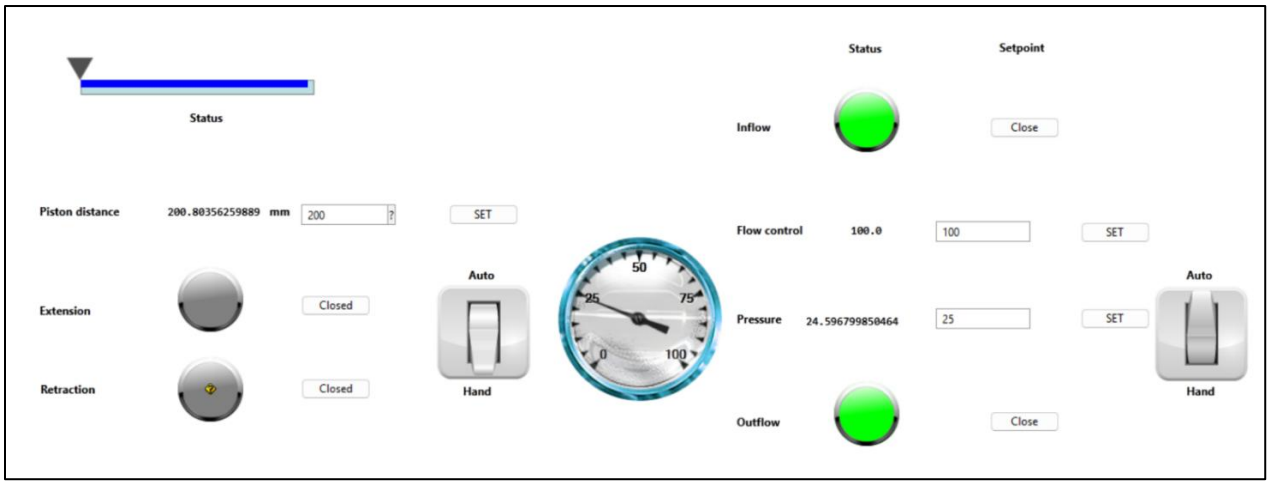



*Figure 7: HMI Developed Using VTScada*

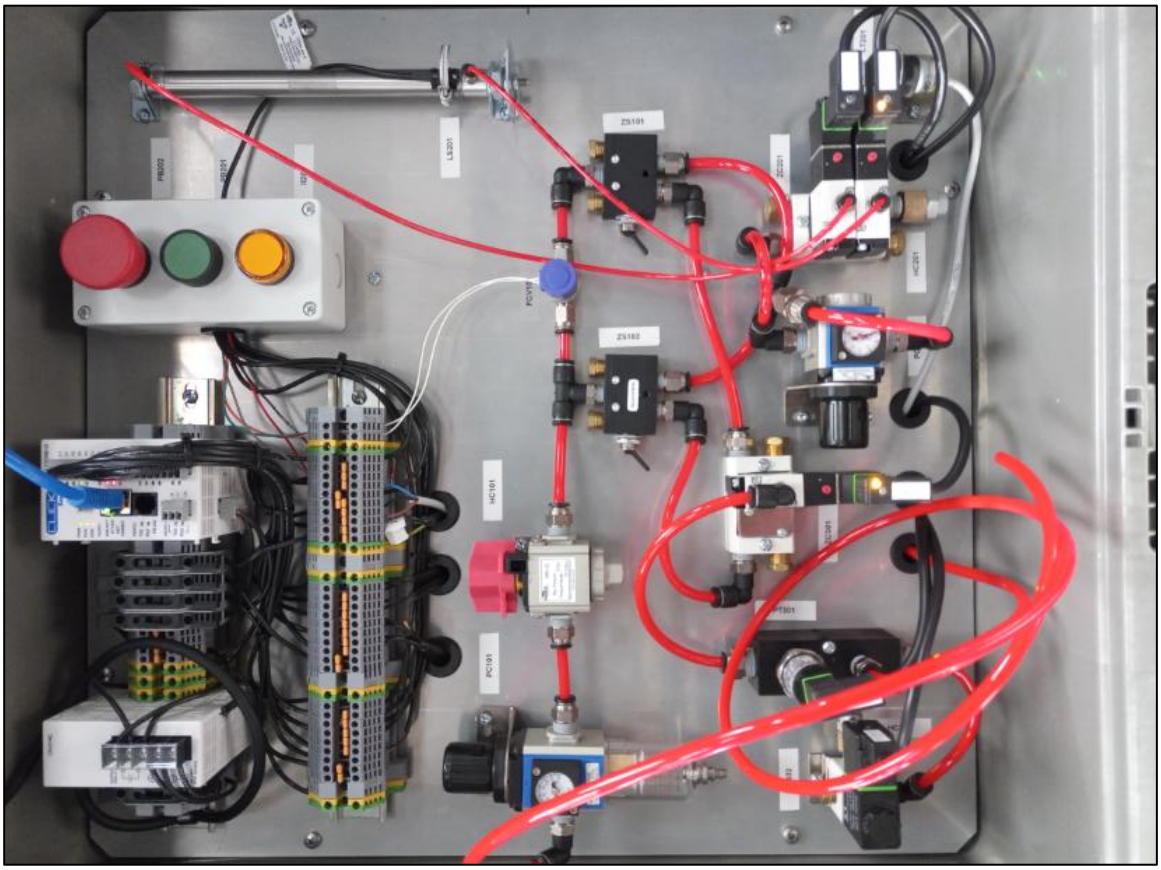


*Figure 8: Industrial Automation Training Kit Used*

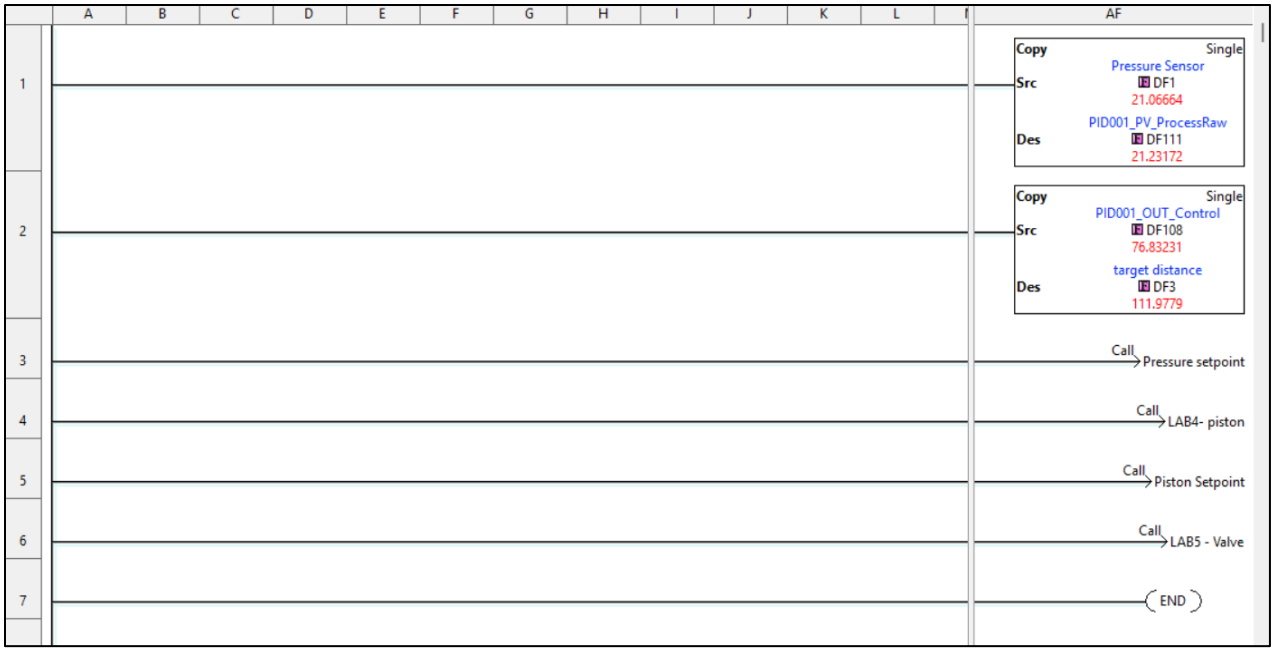



*Figure 9: Main PLC Program Made with ClickPLC*

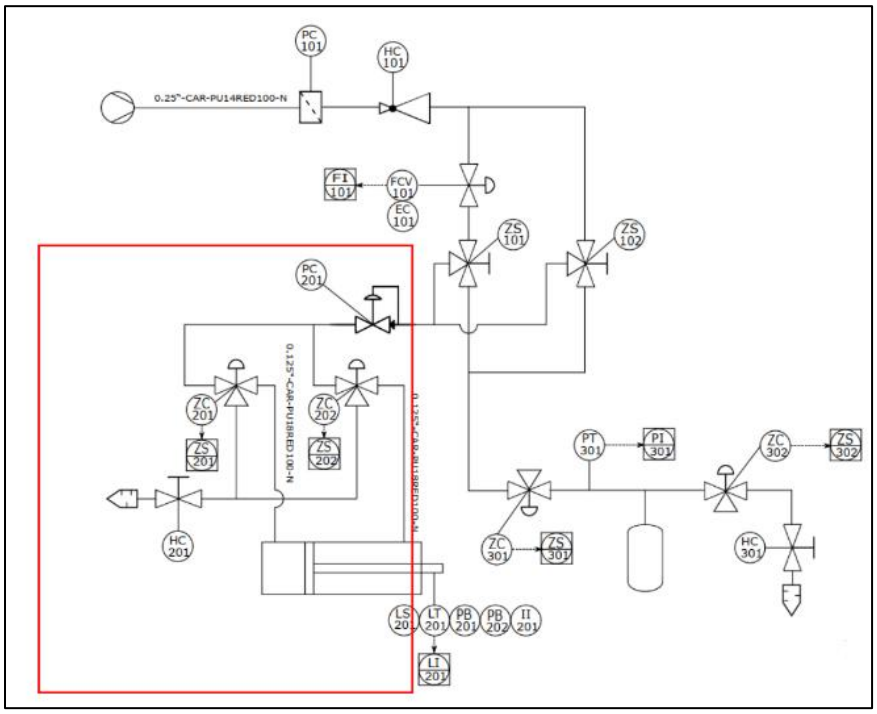



*Figure 10: P&ID of the Lab Kit, Focused on Piston Control*

- **Ultrasonic position transmitter (LT-201):** Measures piston position
- **Pneumatic valve (ZC-201):** Extends piston when actuated
- **Pneumatic valve (ZC-202):** Retracts position when actuated

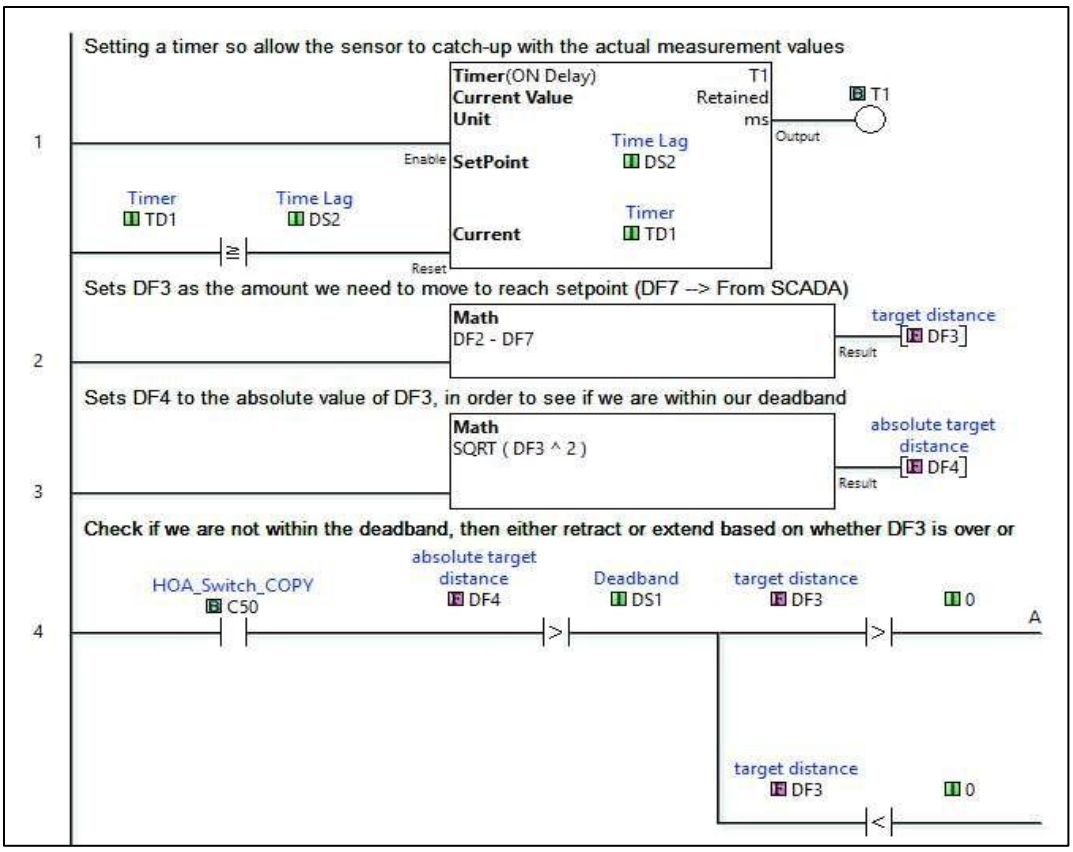



*Figure 11: Piston Subroutine Program (1/2)*

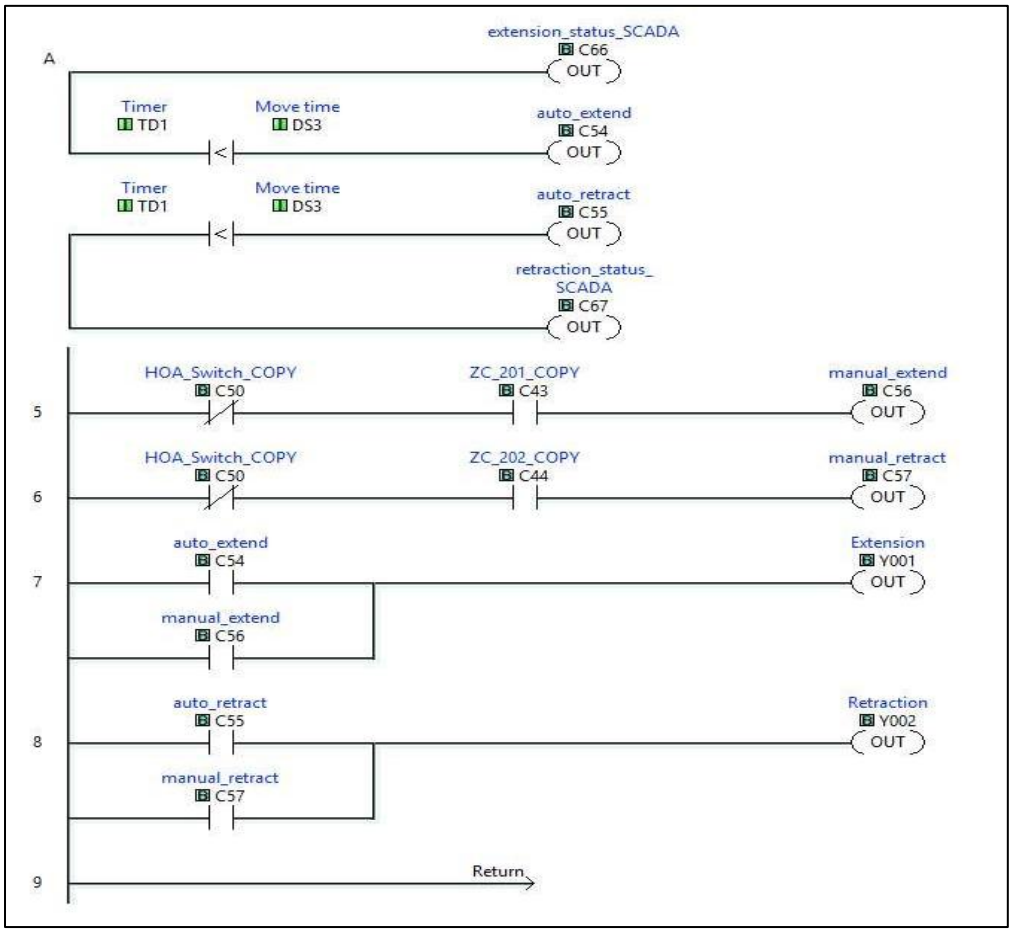



*Figure 12: Piston Subroutine Program (2/2)*

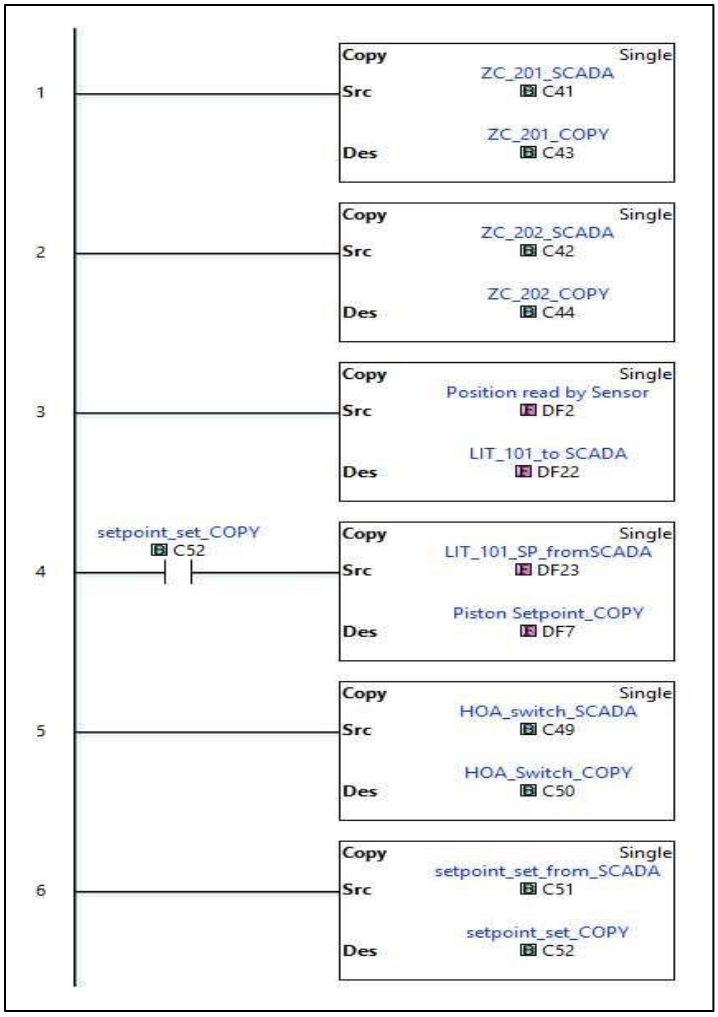



*Figure 13: Piston Setpoint Subroutine Program (1/2)*

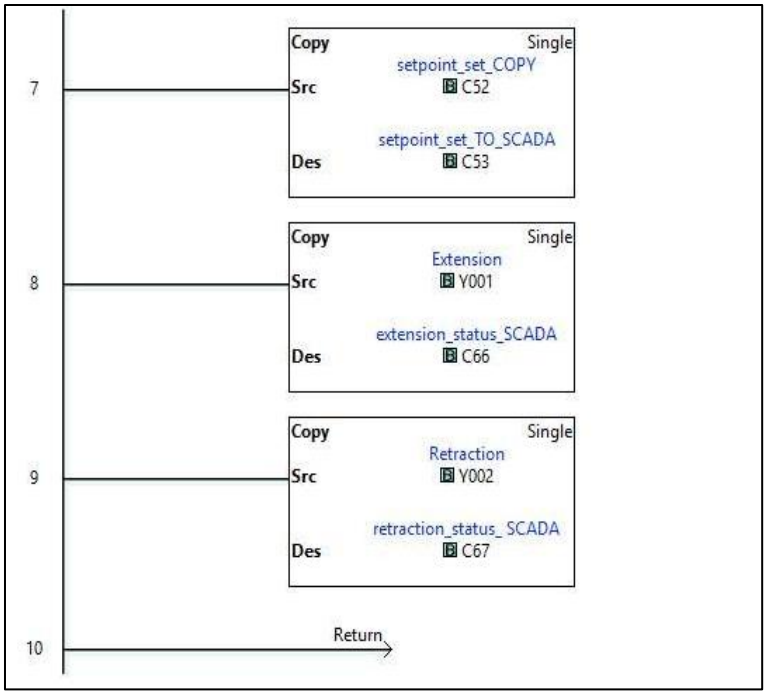



*Figure 14: Piston Setpoint Subroutine Program (2/2)*



*Figure 15: Thermoformed Firetruck Shell*

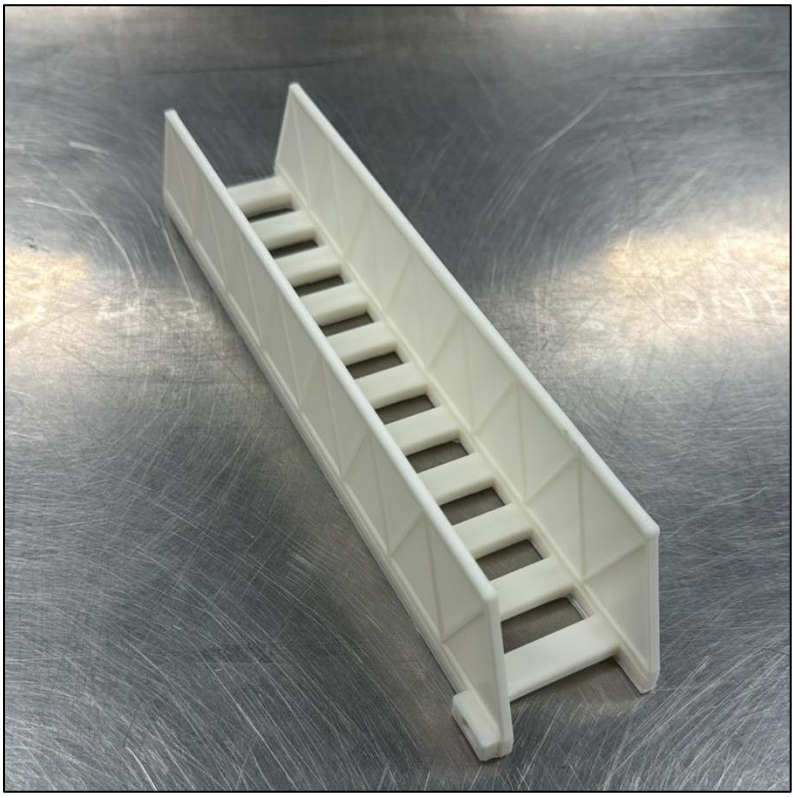


*Figure 16: 3D-Printed Ladder*

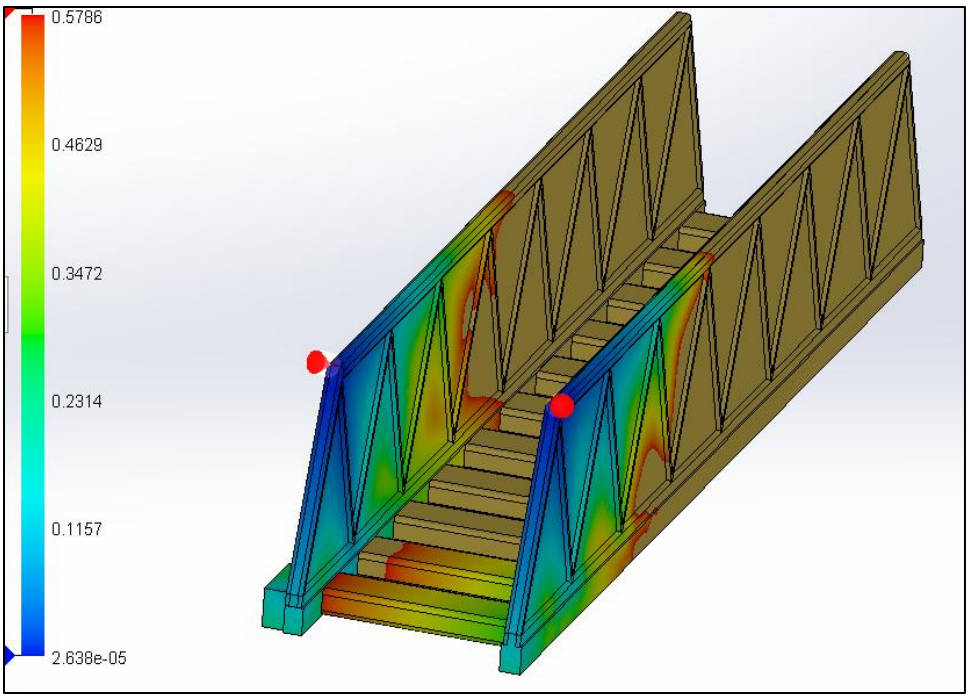



*Figure 17: Injection Molding Simulation at 0.58 s*

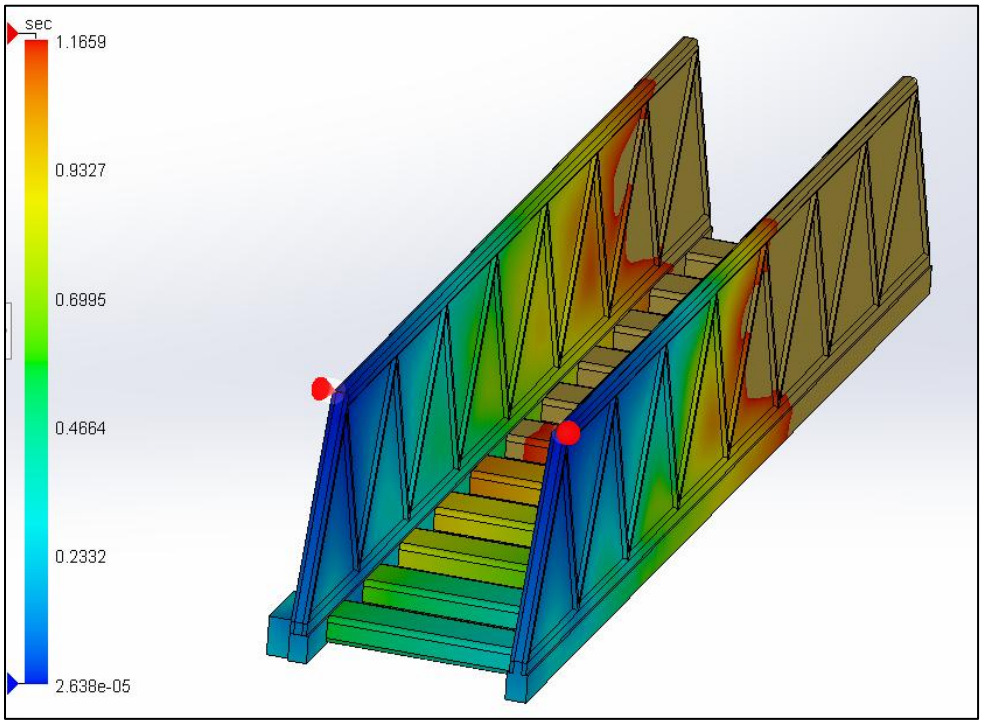



*Figure 18: Injection Molding Simulation at 1.17 s*

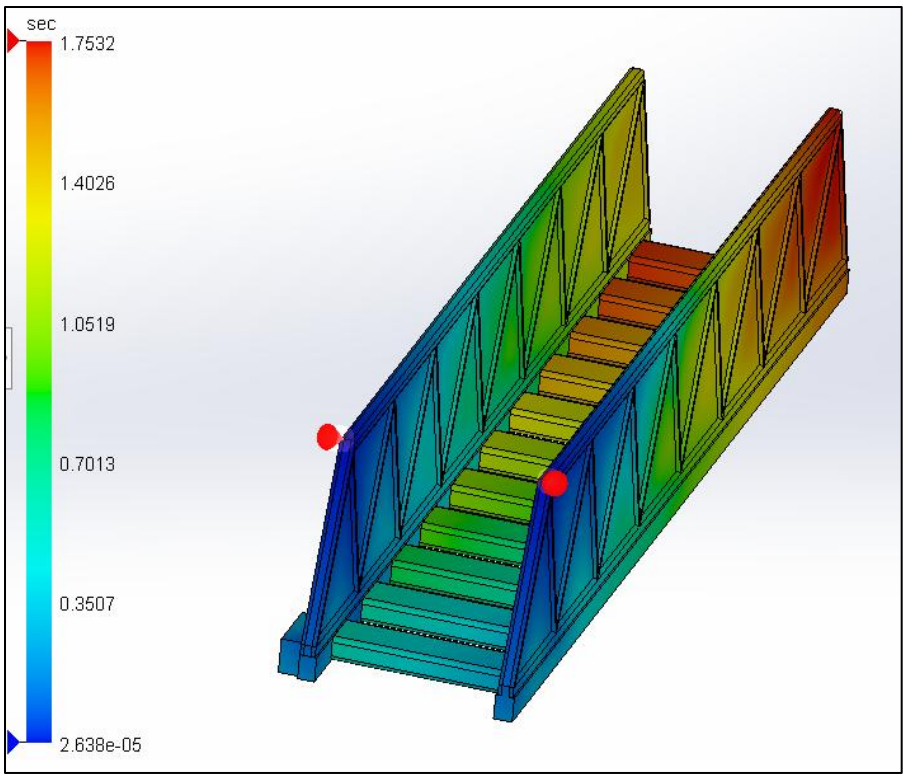



*Figure 19: Injection Molding Simulation at 1.75 s*

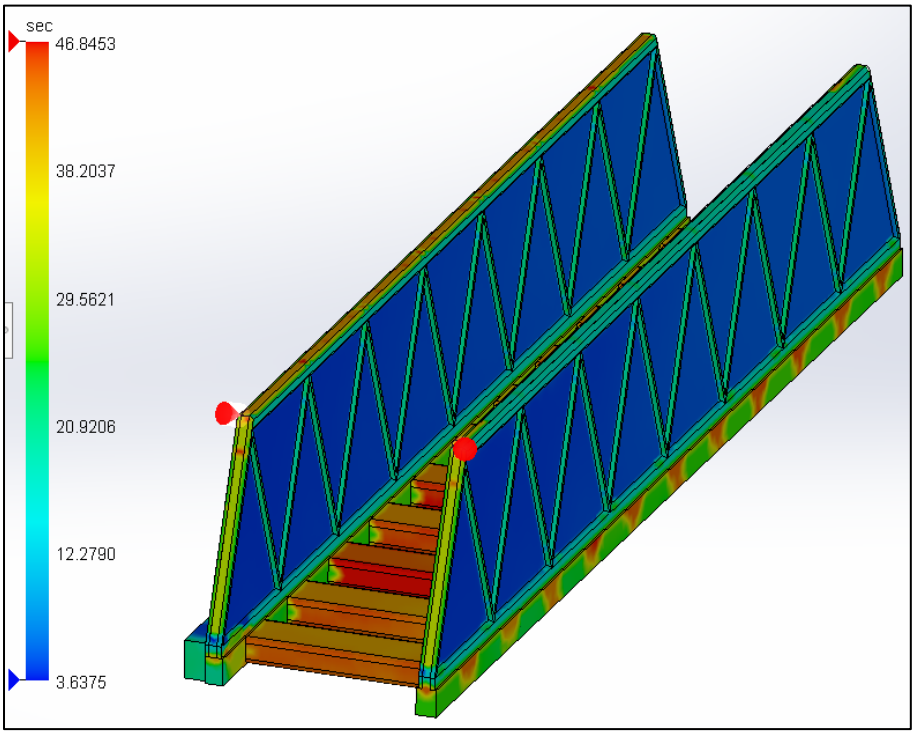



*Figure 20: Cooling Time Analysis from SolidWorks Simulation*

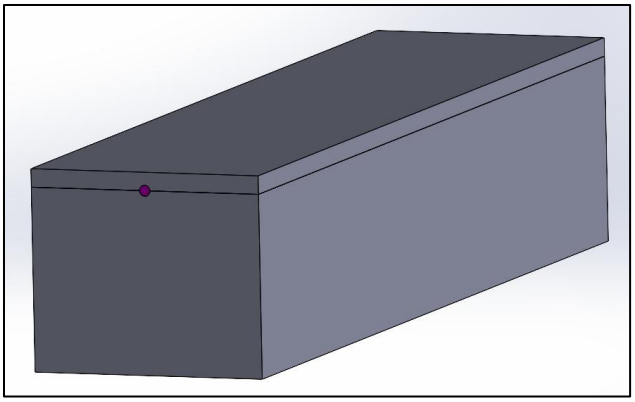


*Figure 21: Closed Ladder Die*

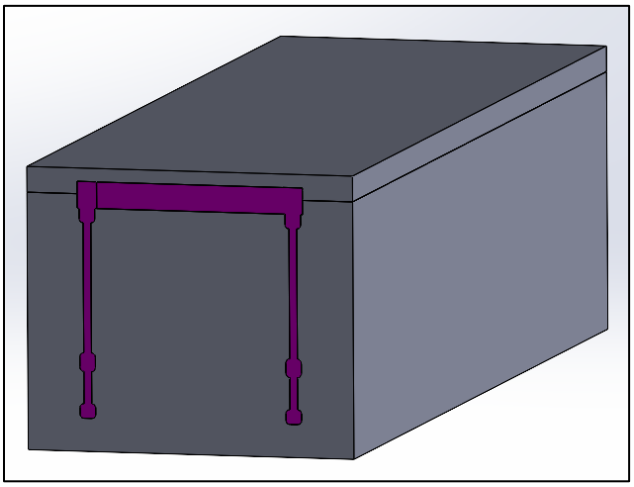


*Figure 22: Section View of Ladder Die*

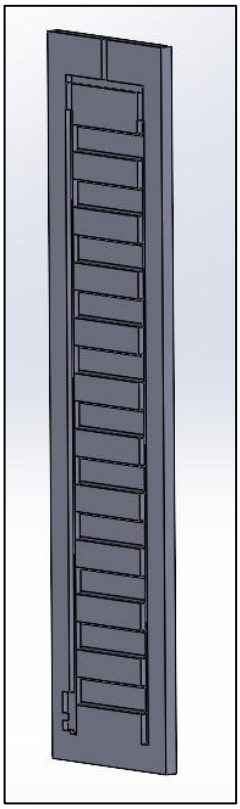


*Figure 23: Top Half of Ladder Die*

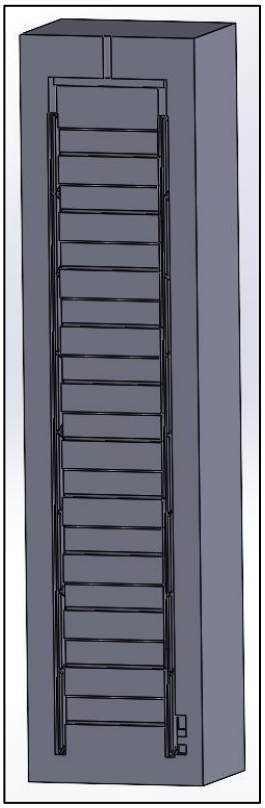


*Figure 24: Bottom Half of Ladder Die*

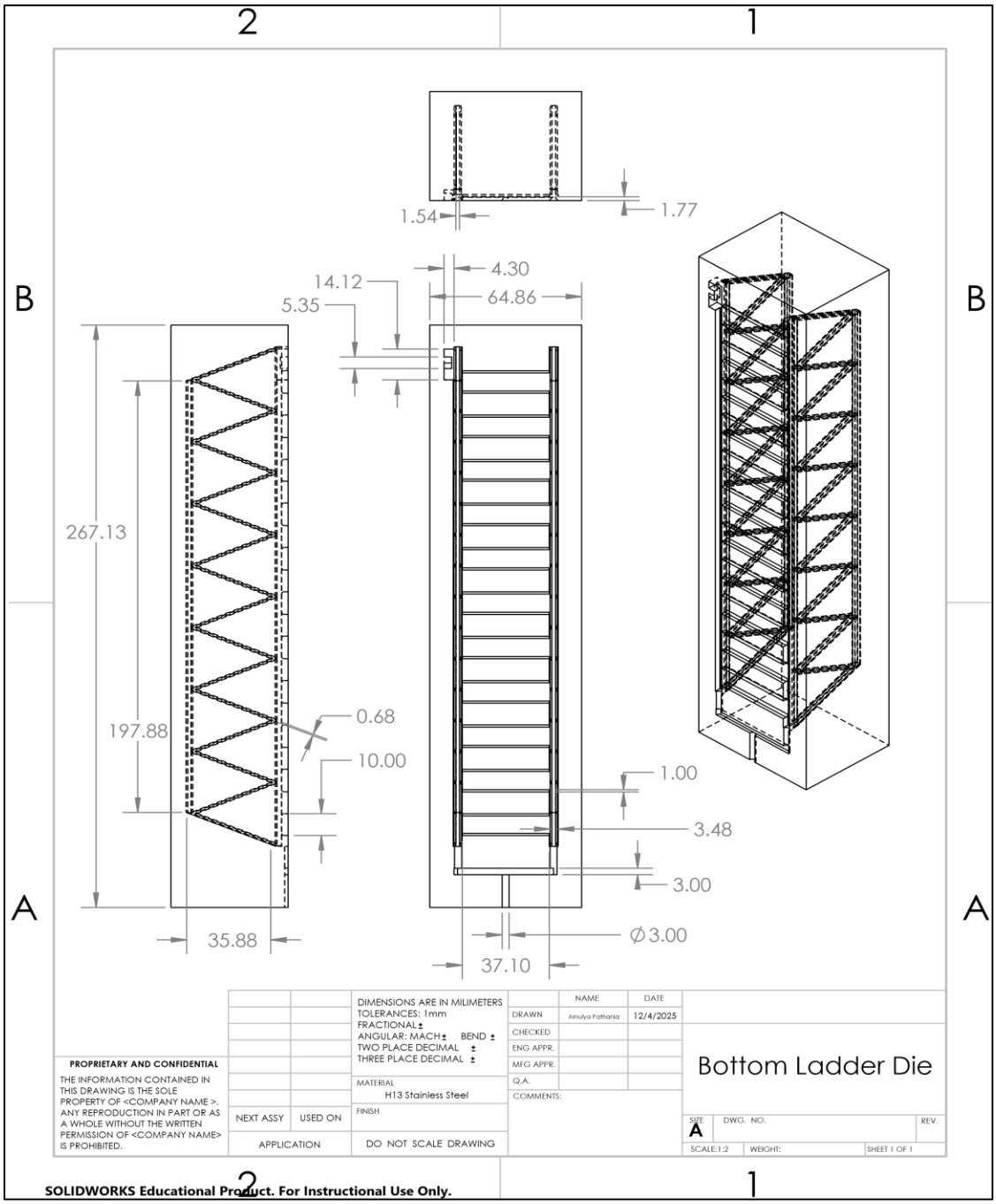



*Figure 25: Engineering Drawing of Bottom Ladder Die*

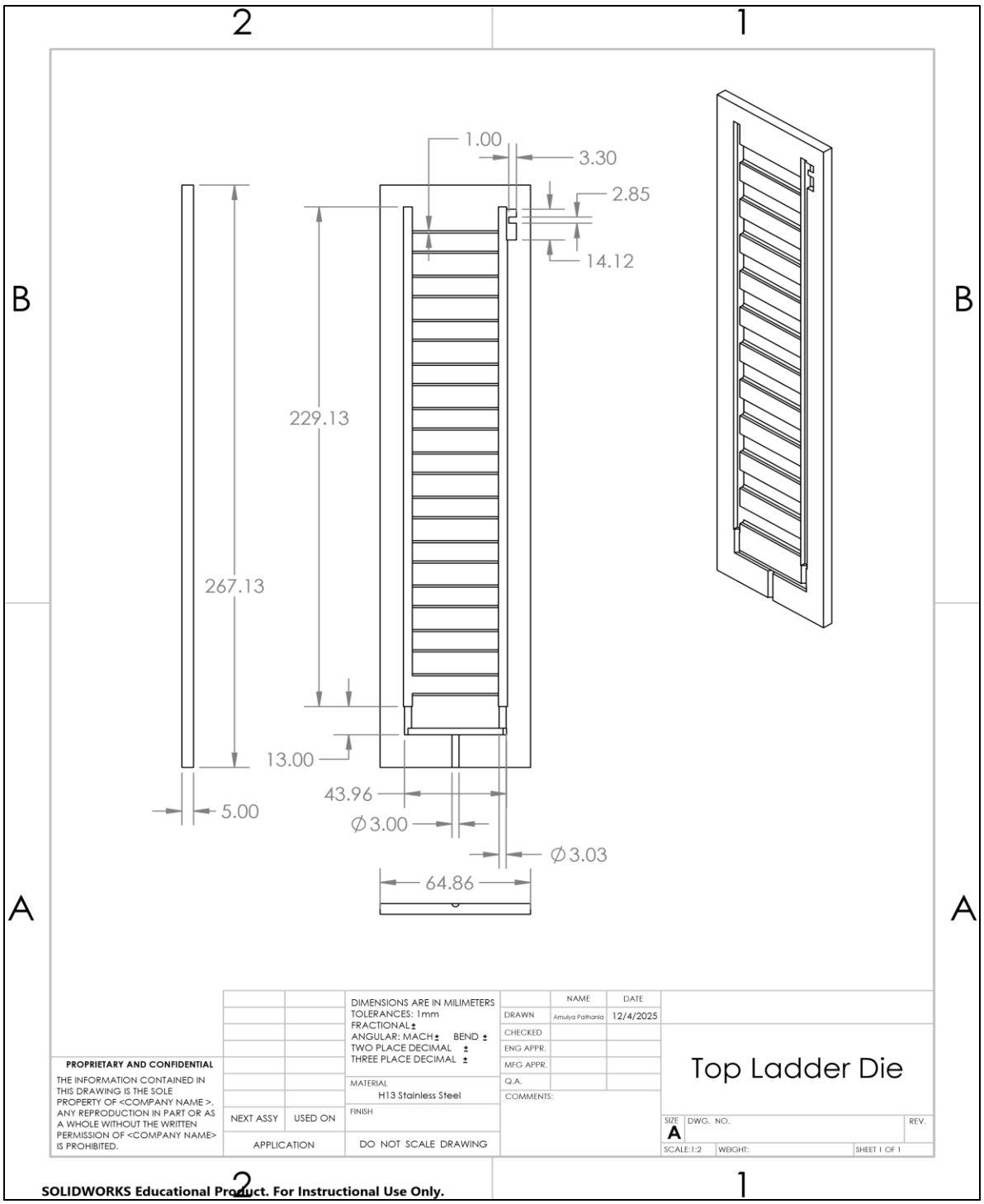



*Figure 26: Engineering Drawing of Top Ladder Die*



*Figure 27: Sand-Cast Aluminum Wheels*

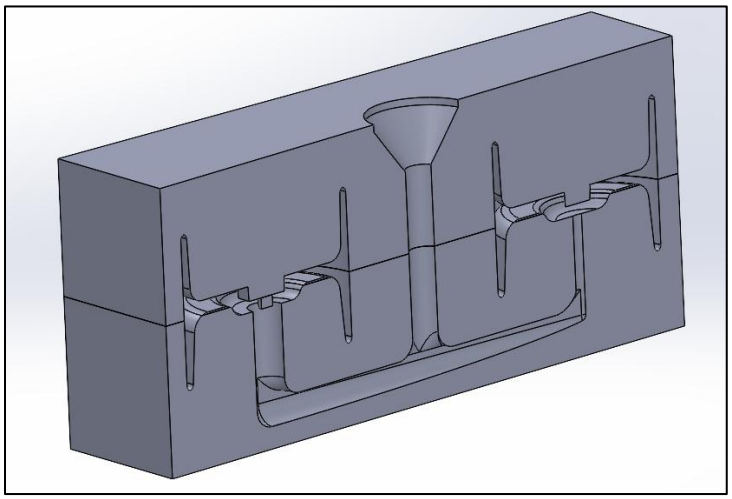


*Figure 28: Section View of Wheel Die*

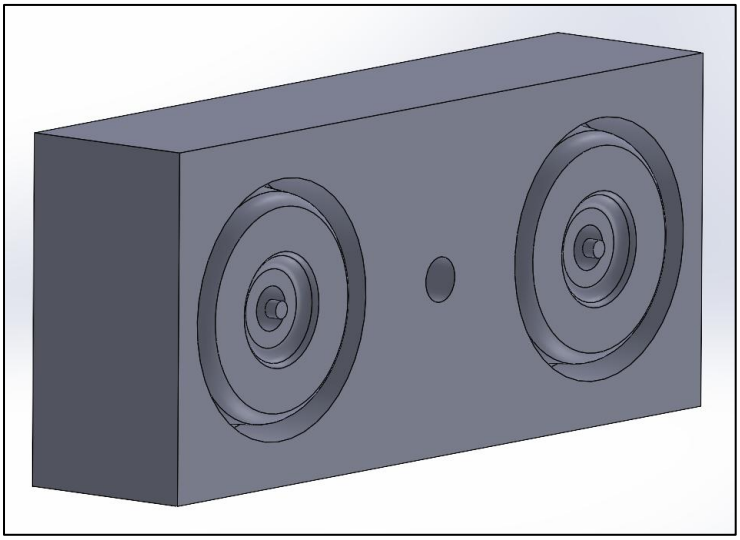


*Figure 29: Bottom Wheel Die*

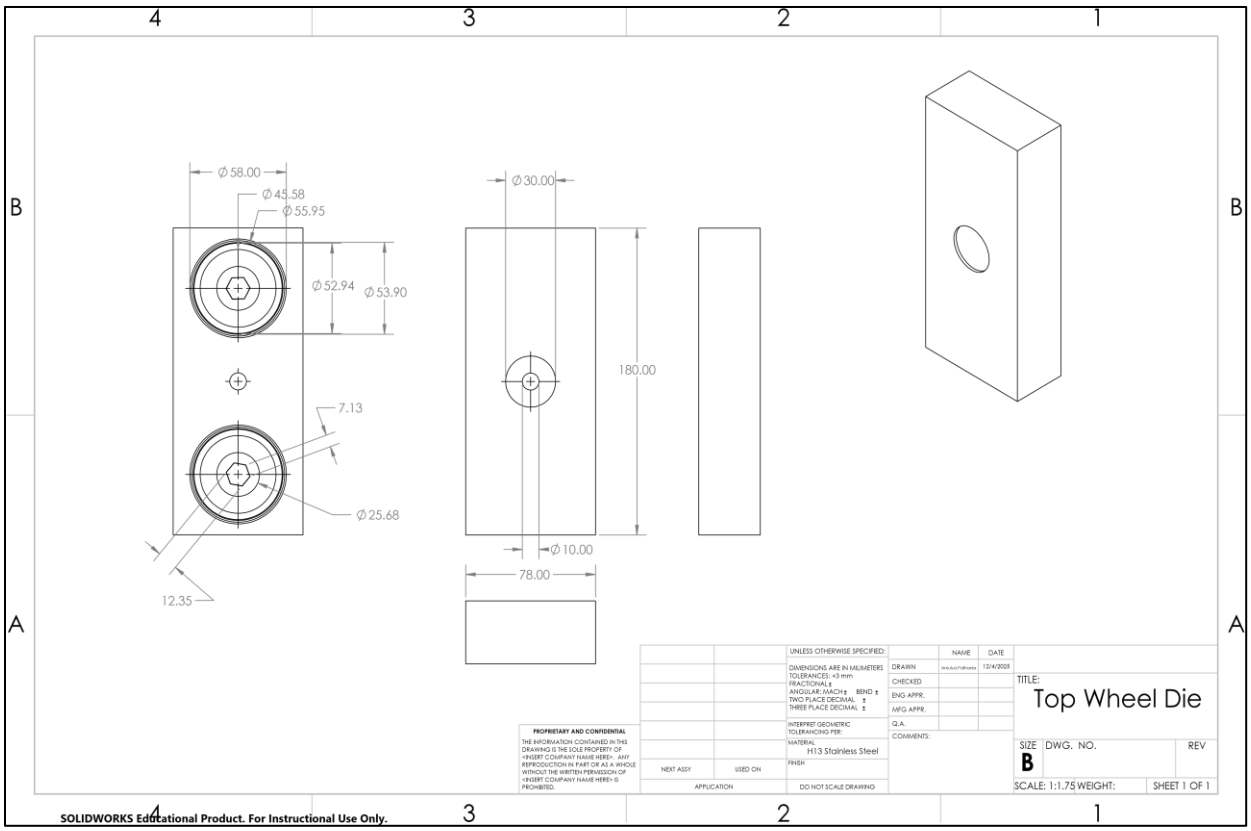



*Figure 30: Engineering Drawing of Top Wheel Die*

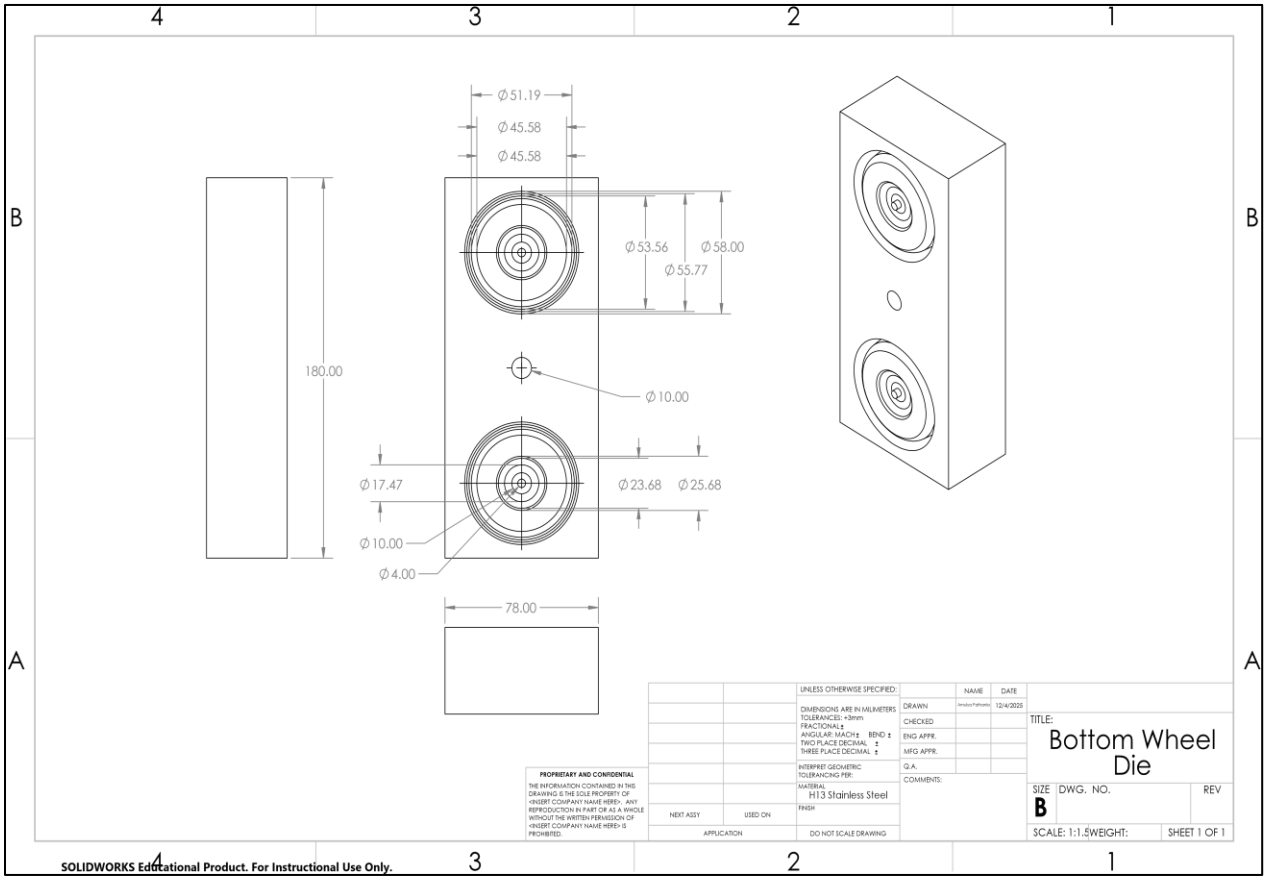



*Figure 31: Engineering Drawing of Bottom Wheel Die*